Cost-Benefit Modeling of Interactive Visual Machine Learning Systems

Yixuan Wang, Rolando Garcia, Alex Endert, and Ross Maciejewski

Abstract—Machine learning systems often benefit from human involvement, but human-in-the-loop support is not free: teams may need to build specialized interfaces, analysts must spend time interacting with them, and some interactions trigger costly retraining or re-optimization. The central design question is therefore not simply whether interaction helps, but when it is worth the added effort relative to other interventions, such as improving the data, revising the model or training specification, or relying on automated workflows alone. In this paper, we build upon Van Wijk's foundational economic model of visualization, extend it by explicitly incorporating machine learning components, and present a cost-benefit framework for reasoning about these choices in interactive visual machine learning systems. We identify measurable quantities that govern practical trade-offs, including analyst time, interaction frequency, training latency, model improvement, downstream operational savings, and failure-prevention value. We organize these quantities into workflow-specific models for three common intervention targets: data iteration, model or specification iteration, and evaluating and understanding model behavior. We then instantiate the framework on representative systems to show how it supports break-even analysis, latency bounds, and stopping criteria for interactive analysis. The contribution is a practical decision framework for reasoning about when interactive visual tooling is likely to be justified, which intervention can provide value, and where human effort can be allocated more effectively across IVML workflows.

Index Terms—Visual analytics for machine learning, interactive visual machine learning, cost-benefit analysis